Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building data acquisition, processing, and distribution pipelines in C++, Java, and Python, using configuration management, build automation, and debuggers, with strong root cause analysis, code reviews, and CI/CD on Linux.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built pipelines for **data acquisition, processing, and distribution** across SQL and cloud services for 500+ users, optimizing performance to cut latency from 1.2s to 350ms.
- Performed **technical root cause analysis** on data integrity issues, implementing corrective actions and validation across 5K+ monthly jobs at 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python data-processing and backend services** handling 10K+ daily requests, building concurrent pipelines and APIs with debuggers and automated testing.
- Contributed to **peer code reviews** and architecture discussions, providing comments and suggestions while reducing turnaround **60%** under high load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built **cross-platform data-processing software** (Java, C++) on Linux, Windows, and macOS using **configuration management, automated build processes, and debuggers** with 20+ test suites.
- Automated **Jenkins CI/CD and nightly build pipelines**, performing root cause analysis on failures and improving pipeline efficiency **68%** in Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **data synchronization and processing** across eight microservices syncing 2M+ records, with REST APIs, real-time updates, and role-based access.
- Automated **build and deployment pipelines** (Jenkins, Docker) with monitoring and logging, cutting deployment time **50%** and enabling rapid issue resolution.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **real-time data pipelines** (Databricks, Spark) and Flask APIs processing 50K+ records in hourly batches, delivering analytics with sub-200ms latency.
- Troubleshot production issues via logging and **root cause analysis**, documenting corrective actions and reducing manual handling **70%**.

Skills

Languages: C++, Java, Python, SQL, JavaScript/TypeScript, Bash
Data Engineering: Data Acquisition, Processing & Distribution, ETL Pipelines, Real-Time Data, Time-Series Data, Data Validation
Software Eng Tools: Configuration Management, Build Automation, Debuggers, Git, Maven, Jenkins
Backend & APIs: Spring Boot, Flask, FastAPI, Node.js, REST APIs, Microservices
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD, Linux, Monitoring
Practices: Code Reviews, Root Cause Analysis, Testing, Agile, Mentoring, Problem-Solving

Projects

CPU Scheduling Simulator Live | *C++, React.js, OS Internals*
- Built a **C++ scheduling simulator** implementing six algorithms with performance analysis and real-time visualization, reducing average turnaround time **25%** across 1K+ scenarios.

Linguistix | *Python, AWS (ECS/Fargate), Docker, CI/CD*

- Developed a **containerized data-processing API** on AWS with Jenkins-style CI/CD, validation checks, and 87% accuracy at production-grade reliability.

Big-Brain Live | *GCP, PostgreSQL, REST APIs, FAISS*

- Built a **data acquisition and processing pipeline** on GCP processing 50K+ records with incremental updates and tuning that cut retrieval latency **40%**.